

Communication Protocol

ISR System
Products

Intelligent pod



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1. Version Description

No.	Version number	Modification item	Author	Date
1.	V1.0	(1) C2M adds online axis calibration on one group/next group/single group confirmation/automatic axis calibration; (2) C2M modifies the online calibration offset type to INT16; (3) C2M adds guidance control instructions and threshold settings; (4) C1M instruction name increase abbreviation; (5) Remote control frame length corrigendum, the entire frame length is 74; (6) Update C2L1, add irradiation related instructions;	MX	20240605
2.	V1.1	(1) Errata RATF; (2) corrigendum of remote control image frame ID; (3) C2M adds automatic tracking instruction;	MX	20240711
3.	V1.2	(1) V1&I1 electronic zoom increases by decimal times;	MX	20240722
4.	V1.3	(1) Correction; (2) Increase the validity of other data in remote control frames; (3) Modify the carrier data flag 0 to be valid; (4) Modify the length field of S2 frame to U16;	MX	20240819
5.	V1.4	(1) C2M adds the calibration command for gyroscope zero-drift;	MX	20240926

6.	V1.5	(1) Added AI enhanced module control instructions; (2) S1 package adds enhanced module status return; (3) S2 package added enhancement module working mode; (4) C1M adds RGIM and SGIM modes; (5) S1 corrigendum; (6) C1M corrigendum;	MX	20241216
7.	V1.6	(1) Added QEC series support description; (2) Corrigendum S1、S2,delete QEC status; (3) AE adjustment ignores the definition of regional parameters and adds small target enhancement/ threat sorting/autonomous confirmation/target detection enhancement, and special application identification; (4) A/AE clearly defines categories;	MX	20250208
8.	V1.7	(1) Add picture-in-picture (PiP) parameter settings to P; (2) Add template matching selection to AE; (3) Add template matching results to S22; (4) Add automatic zoom settings to C2P;;	MX	20250315
9.	V1.8	(1) Add zoom template matching switch parameter input and template matching parameter spatial resolution to C2AE;	WSZ、MX	20250828

		(2) Add a new temperature control switch command to C2M; (3) Add a new S33 telemetry data packet; (4) Add new functions for querying QS0, QS1, and QS2 frames; (5) Add new commands to C2Q, including those for querying laser coding, device information, network information, and laser compensation information; (6) Modify the least significant bit (LSB) of the C2L1 laser coding cycle to 0.01ms, and add a distance compensation command; (7) Add a new tracking fine-tuning command to C2P;		
10.	V1.9	(1) C2E adds functions for frame size setting, chroma setting, and bandwidth adaptive setting. (2) The lower limit of C2E encoding bit rate is changed to 50kbps. (3) The lower limit of C2E encoding frame rate is changed to 1 fps. (4) C2E encoding resolution settings add 144P, 240P, and 320P.	MX	20250922
11.	V1.10	(1) C2MAdd local network settings;	MX	20251030
12.	V1.11	(1) The C2E adds the function of setting coded scene modes; (2) The local network settings of the C2E add a save function;	MX	20251203

		(3) The C2M temperature control adds gear position settings; (4) Correct the definition of S11 encoding resolution;		
13.	V1.12	(1) C2AE:Added single/multi-template configuration; (2) S11:Added connection status indicator; (3) C2P:Added custom OSD configuration; C2Q: Added query commands and QS4 query information frames; (4) C2P:Added mosaic channel configuration for image switching; (5) C2A:Added contrast detection function; (6) C2M:Added log configuration function; (7) S22 Add Target Source Status to Target Category;	MX	20260128
14.	V1.13	(1) Fixed the inconsistency between the QS0 definition and actual implementation. (2) Added S34 threat target status frame to telemetry. (3) Added encryption key function to C2M.	MX	20260418

2. Overview

This document mainly describes the communication protocol content of our company's QD,QP,QG,and QEC series equipment.Unless otherwise specified,the content and transmission method of UART and Ethernet are the same.

3. Physical Layer

3.1. UART

The communication interface is an asynchronous serial port RS422/RS232/TTL, full duplex, with 1 start bit, 8 data bits, 1 stop bit, no check digit, and the baud rate is 115200.

3.2. Ethernet

The default IP address for QD/QP/QG series devices is 192.168.1.100; Remote telemetry communication defaults to UDP service mode, with local port 1030, destination port 4000, and destination address 192.168.1.200; The default video stream address is `rtsp://192.168.1.100:554/stream0`.

The default IP address for QEC series devices is 192.168.1.201; Remote telemetry communication defaults to UDP service mode, with local port 1030, destination port 1040, and destination address 192.168.1.200; The default video stream address is `rtsp://192.168.1.201:554/stream0`.

4. Data Format

Multi-byte data adopts the little-endian mode. The minimum cycle for issuing remote control commands is 20ms. The default cycle for telemetry status return is 20ms.

5. Remote Control Data Frame

The definition of data frames sent by the user to the device is shown in Table 1.

Table 1 remote control data frame

Byte No.	Content	Type	Value and range
1	Frame header 1	UINT8	0x55
2	Frame header 2	UINT8	0xAA
3	Frame length	UINT8	Whole frame data length
4	Frame count	UINT8	Incremental count
5	C1 control instruction package ID	UINT8	Please refer to the table below for detailed definitions
6	C1 control instruction	UINT8	
7	C1 control instruction,the same as above	UINT8	
8	C1 control frame parameters,10 bytes		
9			
10			
11			
12			
13			
14			
15			
16			
17			
18	C2 control the instruction frame ID	UINT8	Please refer to the table below for detailed definitions
19	C2 control instruction	UINT8	
20	C2 control instruction,the same as above	UINT8	
21	C2 control frame parameters,10 bytes		
22			
23			
24			
25			
26			
27			
28			
29			
30			
31	Date		Bit0~6: Year (+2000)
			Bit7~10:Month
32			Bit11~15:Day
33	Time	UINT32	Within milliseconds of the day,LSB=1ms
34			
35			
36			
37	GPS effectiveness	UINT8	Bit0: Latitude, Longitude and Altitude 0: Valid, 1: Invalid

			Bit1:Velocity in the East-North-Up direction 0 is valid,1 is invalid Bit2:Heading angle 0 is valid,1 is invalid Bit3:Pitch, roll angle 0 is valid,1 is invalid Bit4: Other cycle data 0 is valid,1 is invalid Bit5 ~ 7: Reserved
38	Eastward speed of the carrier	INT16	LSB=0.01m/s
39			
40	Northbound speed of the carrier	INT16	LSB=0.01m/s
41			
42	Aerial speed of the carrier	INT16	LSB=0.01m/s
43			
44	Pitch angle of the carrier	INT16	LSB=360/65536°, -90 ~ 90, with zero at horizontal, positive upwards, and negative downwards
45			
46	Roll angle of the carrier	INT16	LSB=360/65536°, -180 ~ 180, with zero at horizontal, right tilt is positive, left tilt is negative
47			
48	Heading angle of the carrier	INT16	LSB=360/65536°, -180 ~ 180, with true north as zero,positive values for north-east, and negative values for north-west.
49			
50	Carrier longitude	INT32	LSB=10 ⁻⁷ °, -180° ~ 180°,WGS-84 -180~180
51			
52			
53			
54	Carrier latitude	INT32	LSB=10 ⁻⁷ °, -90° ~ 90°, WGS-84
55			
56			
57			
58	Carrier height	INT32	LSB=10 ⁻³ m
59			
60			
61			
62	Real time target altitude	INT16	LSB=1m
63			
64	Image frame ID	UINT32	LSB=1
65			
66			
67			
68	Reserved		

~73			
74	Checksum	UINT8	Sum up all the previous bytes and take the lower 8 bits

Note: The attitude adopts the northeast coordinate system, if the external given attitude does not match this coordinate system, it needs to be converted.

5.1. C1#M#ID0x01 System Mode Instruction

C1#M system mode instruction								
No.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do nothing instruction	0x00	-	-	-	-	-	Do not change the servo state
2	Turn off servo OFF	0x01	-	-	-	-	-	Motor stops
3	Collect HIDE	0x02	-	-	-	-	-	
4	Return to center RES	0x03	-	-	-	-	-	Azimuth angle is 0, pitch angle is 0
5	Velocity servo control RATE	0x04	Azimuth angle velocity, INT16, LSB=0.01°/s	Pitch angle velocity, INT16, LSB=0.01°/s	-	-	-	Servo-Driven angle velocity
6	Position servo control GIMF	0x05	Azimuth angle, INT16, LSB=360/65536°	Pitch angle, INT16, LSB=360/65536°	Pitch angle velocity, INT16, LSB=0.01°/s	Pitch angle velocity, INT16, LSB=0.01°/s	-	Servo-Driven frame angle

7	servo spatial angle EULF	0x06	Azimuth angle in space, INT16, LSB=360/65536° True north is 0, and north by east is positive, -180 ~ 180	Pitch angle in space, INT16, LSB=360/65536° Level is 0, upward is positive, -90 ~ 90	-	-	-	Northeast celestial coordinate system
8	Scan SCAN	0x07	Azimuth unilateral amplitude value, the range of UINT16 is 0 ~ 170° LSB=0.01° >180° for azimuth continuous scan	Azimuth velocity value (absolute value), the range of UINT16 is 0 ~ 50°/s LSB=0.01°/s	Pitch side amplitude value, the range of UINT16 is 0° ~ 120° LSB=0.01°	Pitch Velocity value (absolute value), the range of UINT16 is 0 ~ 50°/s LSB=0.01°/s	-	The scanning center angle is the current azimuth angle and pitch angle. If the pitch exceeds the limit, scan according to the maximum range
9	Geographic tracking GEO	0x08	Target longitude INT32, LSB=1e-7°		Target latitude INT32, LSB=1e-7°		Target altitude INT16, LSB=1m	Follow the given geographical location

								n
10	Follow the current spatial angle CEU L	0x09	Azimuth angle velocity, INT16, LSB=0.01°/s	Pitch angle velocity, INT16, LSB=0.01°/s	-	-	-	Keep the spatial angle pointing unchanged (manual lock direction/lock head)
11	Follow the current geographical location CGE O	0x0A	Azimuth angle velocity, INT16, LSB=0.01°/s	Pitch angle velocity, INT16, LSB=0.01°/s	-	-	-	Track current geographic location (manual lock point)
12	Follow the current frame angle CGI M	0x0B	Azimuth angle velocity, INT16, LSB=0.01°/s	Pitch angle velocity, INT16, LSB=0.01°/s	-	-	-	keep the pod at the same relative angle to the nose(manual lock angle/follow)
13	Vertical downward view DOWN	0x0C	-	-	-	-	-	Azimuth 0°, pitch-90°
14	Relative	0x0D	Azimuth angle,	Pitch angle ,	Azimuth angle	Pitch angle velocity,	-	The current

	angle control RGM		INT16, LSB=360/65536°	INT16, LSB=360/65536°	velocity, INT16, LSB=0.01°/s	INT16, LSB=0.01°/s		angle is 0, and azimuth and pitch angles are relative to the current angle
15	Line patrol mode SGM	0x0E	Azimuth angle, INT16, LSB=360/65536°	Pitch angle, INT16, LSB=360/65536°	Azimuth angle velocity, INT16, LSB=0.01°/s	Pitch angle velocity, INT16, LSB=0.01°/s	-	Stable frame angle follow-up
16	Click-to-Track TRA	0x10	Target center azimuth coordinates, UINT16, LSB=1pixel	Target center elevation coordinates, UINT16, LSB=1pixel	Target ID selection, UINT16, LSB=1	-	-	Enter tracking mode with detection targets within a certain range from the center of the target; Target ID selection priority, non-zero is valid
17	Box Selection Tracking CTR	0x11	Target center azimuth coordinate position, UINT16,	Target center pitch coordinate position, UINT16, LSB=1pixel	Target width, UINT16, LSB=1pixel	Target height, UINT16, LSB=1pixel	-	The top left is the zero point, the

	A		LSB=1pixel					bottom right corner is positive, and the range is fixed at 1920 × 1080
18	Auto tracking ATRA	0x12	-	-	-	-	-	Enter tracking mode with the detection target closest to the center of the image
19	Pre-track PTR A	0x13	Target center azimuth coordinate position, UINT16, LSB=1pixel	Target center pitch coordinate position, UINT16, LSB=1pixel	Target width, UINT16, LSB=1pixel	Target height, UINT16, LSB=1pixel	-	The top left is the zero point, the bottom right corner is positive, and the range is fixed at 1920 ×

								1080, platform maintains speed follow-up mode
20	Instruction self-check BIT	0x20	-	-	-	-	-	

5.2. C2#V1#ID0x11 Visible Light Sensor 1 Control command

C2#V1 Visible light sensor 1 control command								
No.	Name	Take value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Continuous zoom+	0x01	Speed value, UINT16, 0: default step; LSB=1, range1~10	-	-	-	-	Field of View -
3	Continuous zoom-	0x02	Speed value, UINT16, 0: default step; LSB=1, range1~10	-	-	-	-	Field of View +
4	Zoom stop	0x03	-	-	-	-	-	
5	Single step zoom+	0x04	Step value, UINT16, 0: default step; 1: the step value is small; 2: the step value is medium; 3: the step value is large;	-	-	-	-	Field of View -

6	Single step zoom-	0x05	Step value, UINT16, 0: default step; 1: the step value is small; 2: the step value is medium; 3: the step value is large;	-	-	-	-	Field of View+
7	Field of view Setting	0x06	UINT16, 0: any value	Field of view value, UINT16, LSB=0.01°	-	-	-	
			UINT16, 1: default value	Default gear, UINT16 0:Wide field of view; 1:medium field of view; 2:small field of view;	-	-	-	
8	Variable Speed setting	0x07	Speed value, UINT16, LSB=0.1, range1~10	-	-	-	-	
9	Focus +	0x10	-	-	-	-	-	
10	Focus -	0x11	-	-	-	-	-	
11	Autofocus	0x12	-	-	-	-	-	
12	Brightness +	0x20	-	-	-	-	-	
13	Brightness-	0x21	-	-	-	-	-	
14	Automatic brightness	0x22	-	-	-	-	-	
15	Contrast +	0x23	-	-	-	-	-	
16	Contrast -	0x24	-	-	-	-	-	
17	Automatic contrast	0x25	-	-	-	-	-	
18	Saturation +	0x26	-	-	-	-	-	
19	Saturation -	0x27	-	-	-	-	-	
20	Automatic Saturation	0x28	-	-	-	-	-	

21	Image enhancement	0x30	Gear value, UINT16, 0: enhance off; 1: enhance 1st gear; 2: enhance 2nd gear; 3: enhance 3rd gear;	-	-	-	-	
22	Near infrared mode	0x31	Mode selection, UINT16, 0: mode off; 1: mode on;	-	-	-	-	
23	Fog Penetration	0x32	Mode selection, UINT16, 0: Fog Penetration off; 1: Fog Penetration on;	-	-	-	-	
24	Electronic zoom	0x40	Integer multiple value, UINT16, 0: X1 (off) ; 1: X2; ... 15: X16;	Decimal multiples, UINT16, LSB=0.1X, range 0~9;	-	-	-	Multiple value = Integer multiple + decimal multiple
25	Electronic image stabilization	0x41	Mode selection, UINT16, 0: image stabilization off; 1: image stabilization on;	-	-	-	-	
26	Power on	0xF0	-	-	-	-	-	
27	Power off	0xF1	-	-	-	-	-	
28	Instruction Transparent Transmission	0xF2	-	-	-	-	-	Transmitting parameters transparent

								rently to the terminal
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5.3. C2#I1#ID0x21 Infrared Sensor 1 Control Command

C2#I1 Infrared sensor control command								
No.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Continuous zoom+	0x01	-	-	-	-	-	Field of View-
3	Continuous zoom-	0x02	-	-	-	-	-	Field of View+
4	Zoom stop	0x03	-	-	-	-	-	
5	Single step zoom+	0x04	Step value, UINT16, 0: default step; 1: the step value is small; 2: the step value is medium; 3: the step value is large;	-	-	-	-	Field of View-
6	Single step zoom-	0x05	Step value, UINT16, 0: default step; 1: the step value is small; 2: the step value is medium; 3: the step value is large;	-	-	-	-	Field of View+
7	Field of view Setting	0x06	UINT16, 0: any value	Field of view value, UINT16,	-	-	-	

				LSB=0.01°				
			UINT16, 1: default value	Default gear, UINT16 0:Wide field of view; 1:medium field of view; 2:small field of view;				
8	Variable Speed setting	0x07	Speed value, UINT16, LSB=0.1, range 1~10;	-	-	-	-	
9	Focus +	0x10	-	-	-	-	-	
10	Focus -	0x11	-	-	-	-	-	
11	Autofocus	0x12	-	-	-	-	-	
12	Brightness +	0x20	-	-	-	-	-	
13	Brightness -	0x21	-	-	-	-	-	
14	Automatic brightness	0x22	-	-	-	-	-	
15	saturation+	0x23	-	-	-	-	-	
16	saturation-	0x24	-	-	-	-	-	
17	Automatic saturation	0x25	-	-	-	-	-	
18	Image enhanceme nt	0x30	Gear value, UINT16, 0: enhance off; 1: enhance 1st gear; 2: enhance 2nd gear; 3: enhance 3rd gear;	-	-	-	-	
19	Electronic zoom	0x40	Integer multiple value, UINT16, 0: X1 (off) ; 1: X2; ... 15: X16;	Decimal multiples, UINT16, LSB=0.1X, range0~9;	-	-	-	Multi ple value = Integer multipl e + decima l multipl e
20	Pseudo color mode	0x50	Mode selection, UINT16, 0: white heat;	-	-	-	-	Based on actual conditi

			1: black heat; 2: Pseudo color mode 1; ... N+1: Pseudo color mode N;					ons
21	Infrared correction	0x60	Mode selection, UINT16, 0: shutter correction; 1: scene correction; 2: pull correction;	-	-	-	-	
22	Power on	0xF0	-	-	-	-	-	
23	Power off	0xF1	-	-	-	-	-	
24	Instruction transparent transmissio n	0xF2	-	-	-	-	-	Transm itting parame ters transpa rently to the termina l

5.4. C2#L1#ID0x31 Laser Control Command

C2#L1 laser sensor 1 control command								
N o.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Single distance measureme nt	0x01	-	-	-	-	-	
3	Continuous distance measureme nt	0x02	Frequency value, UINT16, 0x01:1hz; 0x05:5hz; 0x0A:10hz; 0x0n:nhz;	-	-	-	-	
4	Irradiation	0x03	Code value,	Irradiati on	-	-	-	Please obtain

			UINT16, 0x01:Laser fixed code 1; 0x02:Laser fixed code 2; ... 0x101:Laser custom encoding 1; 0x102:Laser custom encoding2; ...	time, UINT16 ,LSB=1 ms;				the coding value based on the specific model.
5	Distance measureme nt/irradiati on stop	0x04	-	-	-	-	-	
6	Custom en coding settings	0x10	Encoding value, UINT16, 0x101:Laser custom encoding1; 0x102:Laser custom encoding2; ...	Cycle , UINT16 , LSB=0. 01ms;	-	-	-	
7	Distance Compensat ion	0x11	Compensation Channel, INT16, 0x00: Channel 0; 0x01: Channel 1; ... 0x0A: Channel 10;	Compen sation Value, INT16, LSB = 0.1m;	-	-	-	
8	Power on	0xF0	-	-	-	-	-	
9	Power off	0xF1	-	-	-	-	-	
10	Laser enable	0xF2	-	-	-	-	-	Only after enablin g can all laser related comma nds be controll ed
11	Laser disable	0xF3	-	-	-	-	-	
12	Instruction transparent	0xF4	-	-	-	-	-	Transm itting

transmission							parameters transparently to the terminal
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5.5. C2#M#ID0x41 System Debugging Command

C2#M System debugging command								
No.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1.	Do-nothing instruction	0x00	-	-	-	-	-	
2.	Automatic drift compensation	0x01	Mode selection, UINT16, 0: off; 1: on;	-	-	-	-	Can be turned on when in speed following mode, automatically ends after 6 seconds
3.	LMC	0x02	Mode selection, UINT16, 0: off; 1: on;	-	-	-	-	Linear motion compensation
4.	Fine tuning	0x03	Direction selection, UINT16, 0: up; 1: below; 2: left; 3: right;	Adjustment amount: UINT16, LSB=0.001	-	-	-	In the speed following mode, for drift compensation fine-tuning, when the adjustment amount is 0, the default adjustment amount is 0.005 °/s
5.	Online calibration axis	0x10	Mode selection, UINT16, 0: off;	-	-	-	-	Operation must be performed when the axis calibration is activated.

			1: on; 2: parameter saving;							
			3: offset setting;	Horizontal offset pixel, INT16,LSB=1	Horizontal offset pixel, INT16,LSB=1	-	-			
			4:Next group; 5: previous group; 6: single group confirmation; 7: automatic axis calibration	-	-	-	-			
6.	Local Network Setting	0x11	Parameter Type: UINT16 1: IP Address 2: Subnet Mask 3: Gateway	P1	P2	P3	P4	IP Address = P1:P2:P3:P4 Subnet Mask = P1:P2:P3:P4 Gateway = P1:P2:P3:P4 Real-time activation and saving		
			4:MAC Address;	P1	P2	P3	P4	P5	P6	MAC Address: P1:P2:P3:P4:P5:P6
			5:Parameter Saving	-	-	-	-	-	The above settings take effect dynamically, and parameter saving will be performed after a save command is received.	
7.	Log configuration	0x12	Mode Selection, UINT16, 0x01: Export 0xFF: Clear	-	-	-	-			
8.	Encryption Settings	0x13	Mode Selection,	-	-	-	-			

			UINT16, 0x01: Clear Custom;					
9.	Guidance control	0x20	Mode selection, UINT16, 0: off; 1: on;	threshold, UINT16, LSB=1%	-	-	-	Default is off
10.	Automatic tracking	0x30	Mode selection, UINT16, 0: off; 1: on;	-	-	-	-	Default is off
11.	Temperatu re Control	0x50	Mode Selection, UINT16, 0: Off; 1:Defogging 2:De-icing	-	-	-	-	Defaults to on and automatically turns off after 5 minutes.
12.	Gyroscope zero drift calibration	0xE1	-	-	-	-	-	Attention: This operation will automatically turn off the servo and return to the center
13.	Control disable	0xF0	-	-	-	-	-	
14.	Control enable	0xF1	-	-	-	-	-	The following operations are required after the control is enabled
15.	Zero restore factory defaults	0xF2	-	-	-	-	-	
16.	Boresi ght calibr ation	0xF3	0xFFFF	0xFFFF	-	-	-	Set the current angle to zero position
			Azimuth zero position, INT16, LSB=360/65	Pitch Angle zero position, INT16, LSB=360/6	-	-	-	Set the Angle given by the parameter to zero position

			536°	5536				
17.	One click restart	0xF4	-	-	-	-	-	

5.6. C2#Q#ID0x42 System query command

C2#Q system query instructions								
N o.	Name	Value	Parameter byte order					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Null Instruction	0x00	-	-	-	-	-	
2	Query Device Version Information	0x01	-	-	-	-	-	Triggered mode: The telemetry frame of the current cycle will be replaced with the QS0 frame.
3	Query Device Network Information	0x02	-	-	-	-	-	Triggered mode: The telemetry frame of the current cycle will be replaced with the QS1 frame.
4	Query Laser Coding Information	0x03	Coding Value, UINT16, 0x01: Laser Fixed Code 1; 0x02: Laser Fixed Code 2; ... 0x101: Laser Custom Code 1; 0x102: Laser Custom Code 2; ...	-	-	-	-	Triggered mode: The telemetry frame of the current cycle will be replaced with the QS2 frame.
5	Query Laser Compensation Information	0x04	-	-	-	-	-	Triggered mode: The telemetry frame of the current cycle will be replaced with the QS3 frame.
6	Query	0x05	-	-	-	-	-	Trigger mode:

	Custom OSD Information							The telemetry frame of the current cycle is replaced with a QS4 frame.
--	------------------------	--	--	--	--	--	--	--

5.7. C2#P#ID0x51 Image Control Command

C2#P Image control command								
No.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Image channel switch/selection	0x01	Channel number, UINT16, 0x00: visible light 1; 0x01: visible light2; 0x02: infrared 1; 0x03: infrared 2; 0x04: Image Mosaic;	Channel switch, UINT16, 0x00: off; 0x01: on;	-	-	-	The open command is used to turn on video output and switch tracking channels simultaneously
3	OSD display Settings	0x02	Mode selection, UINT16, 0x0: full display; 0x1: fully hidden; 0x2: Boresight crosshair display; 0x3: Boresight crosshair erasure; 0x4: pod angle display; 0x5: pod angle blanking;	-	-	-	-	

			0x6: GPS information display; 0x7: GPS information blanking; 0x8: work mode and other display; 0x9: work mode and other blanking; 0xA: cross display; 0xB: cross blanking; 0xC: Wave gate display; 0xD: Wave gate blanking; 0xE: Detection box display; 0xF: Detection box blanking; 0x10: Character color switch;					
4	Picture-in-Picture Settings	0x03	Mode Setting, UINT16, 0: Off; 1: On; 2: Size Setting;	Sub-picture x, UINT16	Sub-picture y, UINT16	Sub-picture w, UINT16	Sub-picture h, UINT16	
5	Linkage Settings	0x04	Mode setting, UINT16, 0: off; 1: on;	-	-	-	-	After opening, the camera operation is synchronized and the storage instructions are

								synchron ized
6	Automatic Zoom	0x05	Switch Setting, UINT16, 0: Off; 1: On;	Target Pixel Width, UINT16, Range: 100~500	-	-	-	The default setting is Off. When enabled , the camera will automa tically zoom to a range near the target pixel width in trackin g mode.
7	Custom OSD Parameter Configurat ion	0x06	Switch Setting, UINT16 0: Off 1: On	Color Settings; Byte 0: A Byte 1: R Byte 2: G Byte 3: B		Hor izon tal Coo rdin ate, UIN T16	Vertic al Coord inate, UIN T16	Origin (0,0) at the top-left corner, positiv e toward s the bottom -right
8	Custom OSD Character Configurat ion	0x07	UTF-8 Encoded Characters					
9	Tracking Template Setting	0x10	Template Size Setting, UINT16, 0x01: 16*16; 0x02: 32*32; 0x03: 64*64;	-	-	-	-	Default 64*64

			0x04: 128*128;					
10	Tracking Fine Adjustment	0x11	Direction Setting, UINT16, 0: Up; 1: Down; 2: Left; 3: Right;	Step Size, UINT16, LSB = 1 pixel, Range: 0~50	-	-	-	0 is the default value; the default value is 5.
11	Electronic Image Stabilization (EIS)	0x20	Switch Setting, UINT16, 0: Off; 1: On;	-	-	-	-	
12	Infrared Temperature Measurement	0x30	Temperature Measurement Mode, UINT16 0: Off; 1: Single-point Temperature Measurement; 2: Global Temperature Measurement	Target Point Azimuth Coordinates, UINT16 LSB = 1 pixel, valid during single-point temperature measurement	Target Point Pitch Coordinates, UINT16 LSB = 1 pixel, valid during single-point temperature measurement	-	-	Global Display of Maximum and Minimum Temperature Values and Their Positions The top-right corner of the coordinate system is the origin (zero point), and the bottom-right direction is positive.

5.8. C2#E#ID0x61 Encoding Control Command

C2#E Encoding control command								
N o.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Video switch	0x01	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared2; 0x10: all;	Switch, UINT16, 0x00: off; 0x01: on;	-	-	-	
3	Take photo	0x02	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	Mode selection, UINT16, 0x01: single shot;	-	-	-	
				Mode selection, UINT16, 0x02: start continuous shooting;	Interval , UINT16, LSB=1s	-	-	
				Mode selection, UINT16, 0x03: stop continuous;				
4	Encoding the bit rate	0x10	Channel number, UINT16, 0x00: only visible	Bit rate value, UINT16, LSB=1kbps,	-	-	-	

	setting		light 1; 0x01: only visible light 2; 0x02: only infrared1; 0x03: only infrared2; 0x10: all;	Range 50~20000				
5	Encoding frame rate Settings	0x11	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	Frame rate value, UINT16, LSB=1fps, Range 1~50	-	-	-	
6	Encoding resolution Settings	0x12	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	Resolution value, UINT16, 0:primitive; 1:640*480/480P; 2:1280*720/720P; 3:1920*1080/1080P ; 4: 2560*1440; 5: 3840*2160; 6:192*144/144P; 7:320*240/240P; 8:480*360/360P;	-	-	-	
7	Encoding format	0x13	Channel number, UINT16,	Format , UINT16, 0x00: H264;	-	-	-	

	Settings		0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	0x01: H265;				
8	Encoding GOP settings	0x14	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	GOP value , UINT16, LSB=1, Range : 10~255	-	-	-	
9	Video encoding scene settings	0x15	Scene mode, UINT16, 0: Balanced mode; 1: Ultra HD; 2: HD; 3: Ultra smooth; 4: Smooth;	-	-	-	-	Default : Balanced mode
10	Frame Size Setting	0x20	Channel Number, UINT16, 0x00: Visible light 1 only; 0x01: Visible light 2 only; 0x02: Infrared 1 only;	Frame Size: UINT16 0x00: Full Frame 0x01: 3/4 Frame 0x02: 1/2 Frame 0x03: 1/4 Frame	-	-	-	

			0x03: Infrared 2 only; 0x10: All;					
11	Chroma Setting	0x21	Channel Number: UINT16 0x00: Visible light 1 only 0x01: Visible light 2 only 0x02: Infrared 1 only 0x03: Infrared 2 only 0x10: All	Chroma Selection: UINT16 0x00: Full Color 0x01: Grayscale	-	-	-	
12	Bandwidth Adaptive	0x22	Switch: UINT16 0x00: Off 0x01: On	-	-	-	-	Adaptive Bit Rate: Default is Off
13	Control disabled	0xF0	-	-	-	-	-	
14	Control enable	0xF1	-	-	-	-	-	The following operations are required after the control is enabled
15	Storage Formatting	0xF2	Channel number, UINT16, 0x00: only visible light 1;	-	-	-	-	

			0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;					
16	Encoding parameter save	0xF3	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	-	-	-	-	
17	Encoding parameters revert to default	0xF4	Channel number, UINT16, 0x00: only visible light 1; 0x01: only visible light 2; 0x02: only infrared 1; 0x03: only infrared 2; 0x10: all;	-	-	-	-	

5.9. C2#A#ID0x71 AI Control command

C2#A AI control command								
N o.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Target detection switch	0x01	Category switch, UINT16, 0off, 1on; Bit0: Aircraft; Bit1: boat; Bit2: car; Bit3: human; ...	-	-	-	-	
3	Detect threshold settings	0x02	threshold, UINT16, LSB=0.01;	-	-	-	-	
4	Enable detection data	0x03	Enable switch, UINT16, 0off, 1on;	-	-	-	-	After enabling, telemetry data adds status frame2, which alternates with frame 1 and returns at 25Hz each
5	Contrast Detection	0x10	Switch, UINT16, 0 : Off, 1 : On;	-	-	-	-	

5.10. C2#AE#ID0x81 AI Enhanced Control command

C2#AE AI enhanced control command								
N o.	Name	Value	Parameter byte number					Notes
			1~2	3~4	5~6	7~8	9~10	
1	Do-nothing instruction	0x00	-	-	-	-	-	
2	Target detection switch	0x01	Category switch, UINT16, 0off, 1on; Bit0: Aircraft; Bit1: boat; Bit2: car; Bit3: human; ...	-	-	-	-	
3	Template matching switch	0x02	UINT16 0x00: Off 0x01: On Bits 1~2: Mode 0x00: Single Template 0x01: Multi Template Bits 3~7: Reserved Bits 8 ~ 15: Spatial Resolution, LSB = 0.1 (Value Range: 0 ~ 255)	Target Coordinates x, UINT16	Target Coordinates y, UINT16	Target Width w, UINT16	Target Height h, UINT16	When the width and height of the target coordinates are both zero, the last input coordinate position is used, and the template is not updated; when the coordinates are valid,

								the template is updated.
4	Everything recognition switch	0x03	UINT16 0x00:off 0x01:on	-	-	-	-	
5	Multi-target tracking switch	0x04	UINT16 0x00:off 0x01:on	-	-	-	-	
6	ISAS(Obstacle avoidance at sea)switch	0x05	UINT16 0x00:off 0x01:on	-	-	-	-	
7	Zoom target detection switch	0x11	Category switch, UINT16, 0off, 1on; Bit0:Aircraft; Bit1: boat; Bit2: car; Bit3: human; ...	-	-	-	-	
8	Zoom Template Matching Switch	0x12	UINT16 0x00: Off 0x01: On Bits 1~2: Mode 0x00: Single Template 0x01: Multi Template Bits 3~7: Reserved Bits 8~15: Spatial Resolution, LSB = 0.1 (Value Range: 0~255)	Target Coordinates x, UINT16	Target Coordinates y, UINT16	Target Width w, UINT16	Target Height h, UINT16	When both the width and height of the target coordinates are zero, the last input coordinate position is used and

								the template is not updated; when the coordinates are valid, the template is updated.
9	Multiplier Everything identification switch	0x13	UINT16 0x00:off 0x01:on	-	-	-	-	
10	Reset ISAS tracking	0x21	-	-	-	-	-	Special Application 1
11	Set ISAS ignore area	0x22	UINT16 0x00:off 0x01:on	UINT16 Top-Left Vertex Coordinate X	UINT16 Top-Left Vertex Coordinate Y	UINT16 Bottom-Left Vertex Coordinate X	UINT16 Bottom-Left Vertex Coordinate Y	Special Application 1
12	Small target enhancement switch	0x23	UINT16 0x00:off 0x01:on	-	-	-	-	
13	Set threat ranking strategy	0x24	Threat ranking strategy: 0x00:none 0x01:Confidence priority 0x02:Area priority 0x03:ID priority	-	-	-	-	

			0x04:Category sorting					
14	Confirmation of autonomous control process	0x30	UINT16, 0x00: Waiting for confirmation 0x01: Enter the next process 0x02: Return to initial process	-	-	-	-	
15	Target Detection Enhanced	0xA0	UINT16 0x00:off 0x01:on	-	-	-	-	Special Application 2

6. Telemetry Status Frames

6.1. Telemetry Status Frames S11

Table 2 telemetry status frames S11

Byte No.	Content	Type	Value and range
1	Frame header 1	UINT8	0xAA
2	Frame header 2	UINT8	0x55
3	Frame length	UINT8	Whole frame data length
4	Frame identification code	UINT8	0x11: System status frame
5	System Mode	UINT8	See C1#M package definition for details
6	Debugging status	UINT8	Bit0: Drift compensation 0 off, 1 on
			Bit1: LMC 0 off, 1 on
			Bit2: Fine tuning instruction 0 none, 1 successful
			Bit3: Calibration axis 0 off, 1 on
			Bit4: Target calibration command 0 none, 1 successful
			Bit5~Bit7: reserved

7	Servo fault code	UINT8	Bit0: Platform self-test 0 no faults, 1 faults
			Bit1: Azimuth encoder 0 no faults, 1 faults
			Bit2: Pitch encoder 0 no faults, 1 faults
			Bit3: roll encoder 0 no faults, 1 faults
			Bit4: Gyroscope 0 no faults, 1 faults
			Bit5: Platform communication 0 no faults, 1 faults
			Bit6: External square encoder 0 no faults, 1 faults (two-axis four-frame)
			Bit7: External pitch encoder 0 no faults, 1 faults (two-axis four-frame)
8	Load status	UINT8	Bit0: Visible light sensor 1, 0 no faults, 1 faults
			Bit1: Visible light sensor 2, 0 no faults, 1 faults
			Bit2: Infrared sensor, 0 no faults, 1 faults
			Bit3: Laser sensor, 0 no faults, 1 faults
			Bit4: Visible light sensor 1 power supply, 0 not powered on, 1 powered on
			Bit5: Visible light sensor 2 power supply, 0 not powered on, 1 powered on
			Bit6: Infrared sensor 1 power supply, 0 not powered on, 1 powered on
			Bit7: Laser sensor 1 power supply, 0 not powered on, 1 powered on
9	Pod azimuth angle	INT16	LSB=360/65536°, -180° ~ 180°
10			
11	Pitch Angle of the pod	INT16	LSB=360/65536°, -120° ~ +90°
12			
13	Pod roll Angle	INT16	LSB=360/65536°, -180° ~ 180°
14			
15	Azimuth Angle	INT16	LSB=0.01°/s

16	velocity		
17	Pitch Angle	INT16	LSB=0.01°/s
18	velocity		
19	Roll Angle	INT16	LSB=0.01°/s
20	velocity		
21	Pod pitch attitude Angle	INT16	LSB=360/65536°, -90° ~ 90°, level is zero, top is positive, bottom is negative
22			
23	Pod roll attitude Angle	INT16	LSB=360/65536°, -180° ~ 180°, level is zero, right tilt is positive, left tilt is negative
24			
25	Pod azimuth attitude Angle	INT16	LSB=360/65536°, -180° ~ 180°, true north is zero, north by east is positive, north by west is negative
26			
27	Geographic location target longitude	INT32	LSB=10 ⁻⁷ °
28			
29			
30			
31	Geographic location target latitude	INT32	LSB=10 ⁻⁷ °
32			
33			
34			
35	Geographic location target height	INT16	LSB=1m
36			
37	Laser status	UINT8	Bit0: Enable state; 0x00: not enabled; 0x01: enabled;
			Bit1~3: Working mode; 0x00: Stop; 0x01: Single distance measurement; 0x02: 1hz continuous ranging; 0x03: 5hz continuous ranging; 0x04: Laser irradiation;
			Bit4~7: Laser coding; 0x00: Invalid; 0x01: Code 1; ...
		UINT8	Fault code, defined as actual;
		UINT16	Distance value; LSB=1m;
40			
41	Laser replenishment	UINT16	Bit0~3: Distance value decimal place

42			LSB=0.1m; Bit4: range valid mark; 0x00: Invalid 0x01: Valid Bit5: Range count; Bit6~9 : range delay compensation , LSB=10ms; Bit10~15: standby;
43	Track status	UINT8	Bit0~1: Tracking status; 0x00: Stop; 0x01: Lost; 0x02: Tracking; 0x03: Pre tracking;
			Bit2~4: Template size status; 0x01: 16*16; 0x02: 32*32; 0x03: 64*64; 0x04: 128*128;
			Bit5~7: Standby
44		UINT16	Target center horizontal coordinate;
45			LSB=1pixel, The top left is zero,and the bottom right is positive;
46		UINT16	Target center vertical coordinate ;
47			LSB=1pixel, The top left is zero,and the bottom right is positive;
48		UINT16	Target width; LSB=1pixel
49			Target height; LSB=1pixel
50			
51			
52	Standby	-	-
53			
54	Current Main Image Sensor Status	UINT8	Bit0~3: Sensor channel 0x00: Visible light sensor 1; 0x01: Visible light sensor 2; 0x02: Infrared sensor 1;

			0x03: Infrared sensor 2;
			Bit4~7: Electronic zoom state; 0: X1 (off) ; 1: X2; ... 15: X16;
55		UINT8	Bit0~1: Image enhancement mode; 0x00: Enhancement level; 0x01: Enhanced 1st gear; 0x02: Enhanced 2nd gear; 0x03: Enhanced 3rd gear;
			Bit2: Dehazing mode; 0x00: Dehazing off; 0x01: Dehazing on;
			Bit3: Near infrared mode; 0x00: Near infrared off; 0x01: Near infrared on;
			Bit4: Stabilization Mode; 0x00: Stabilization off; 0x01: Stabilization on;
			Bit5~6: Field of view zoom state; 0x00: stop changing; 0x01: Field of view is zooming in; 0x02: Field of view is zooming out;
			Bit7: Pseudo color mode; 0x00:Pseudo color off; 0x01: Pseudo color on;
56		UINT16	Magnification factor of view Angle; LSB=0.1times;
57			
58		UINT16	Horizontal field of view Angle; LSB=0.01°;
59			
60		UINT16	Vertical field of view Angle; LSB=0.01°;
61			
62	Image status	UINT8	Bit0:Electronic image stabilization switch;

			0x0:Off; 0x01:On;
			Bit1:Picture in picture switch; 0x00: Off; 0x01: On;
			Bit2: Object detection switch; 0x00: Off; 0x01: On;
			Bit 3: Connection Status; 0x00: Disconnected; 0x01: Connected;
			Bit 4: Reserved;
			Bit5~7: standby;
63	Standby	UINT8	-
64	Encoding(alternating feedback when multiple sensors are used)	UINT8	Bit0~3:Sensor channel 0x00: Visible light sensor 1; 0x01: Visible light sensor 2; 0x02:Infrared sensor 1; 0x03:Infrared sensor 2;
			Bit4~6:Video resolution; 0: 1920*1080/1080P; 1: 640*480/480P; 2: 1280*720/720P; 3: 2560*1440/2K; 4: 3840*2160/4K; 5: 192*144/144P; 6: 320*240/240P; 7: 480*360/360P;
			Bit7: Encoding format; 0x00: H264; 0x01: H265;
65		UINT8	Bit0~6:Photography count value,LSB=1 Bit7: Recording status 0x00: off; 0x01: on;
66		UINT16	Bit0~14 storage capacity,LSB=0.1GB
67			Bit15: Capacity marker;

			0x00: Remaining capacity; 0x01: Total capacity;
68		UINT16	Video stream,LSB=1kbps
69			
70			
71			
72	System temperature	INT8	LSB=1°C,-128 ~ 127
73	Checksum	UINT8	Sum all the previous bytes and take the lower eight bits
Note: QD/QP/QG series: When telemetry status frame 2 is not enabled, the telemetry data frequency is 50hz, and the frame 1 refresh rate is 50hz;When telemetry status frame 2 is enabled,the telemetry data frequency is 50hz, frame 1 refresh rate is 25hz,the frame 2 refresh rate is 25hz, alternately returning;The default state is not enabled. QEC Series: When connected to pod equipment, the telemetry data frequency is 50hz,the frame 1 refresh rate is 25hz, and the frame 2 refresh rate is 25hz, alternately returning; When the pod equipment is not connected, the telemetry data frequency is 50hz, the frame 1 refresh rate is 0hz, and frame 2 refresh rate is 50hz.			

6.2. Telemetry status FrameS22

Table 3 Telemetry status frameS22

Byte No.	Content		Type	Value and range
1	Frame header1		UINT8	0xAA
2	Frame header2		UINT8	0x55
3	Frame length		UINT16	Whole frame data length
4				
5	Frame identification code		UINT8	0x22, System status frame
6	Image frame ID		UINT32	Image frame ID
7				
8				
9				
10	Time stamp		UINT64	The Current external timing time stamp , LSB=1ms
11				
12				
13				
14				
15				
16				
17				
18	Target number		UINT8	The maximum number is32
19	Goal	Category	UINT8	Bit 0 ~ 3:Target Category

	1			0x00: Aircraft; 0x01: Ship; 0x02: Car; 0x03: Person; Bit 4 ~ 6:Target Source 0x00: Main View; 0x01: TV1; 0x02: IR1; 0x03: TV2; 0x04: IR2; 0xff: Template Matching Result;
20		ID	UINT8	LSB=1
21		Confidence level	UINT8	LSB=0.00390625
22		Horizontal coordinates	UINT16	LSB=1pixel , The top left is zero,and the bottom right is positive
23				
24		Vertical coordinates	UINT16	LSB=1pixel , The top left is zero,and the bottom right is positive
25				
26		Width	UINT16	LSB=1pixel
27				
28				
29	Height	UINT16	LSB=1pixel	
	...			
	Target n			
19+n*11	Frame end 1	UINT8	0xAA	
20+n*11	Frame end 2	UINT8	0x55	
21+n*11	Checksum	UINT8	Sum all the previous bytes and take the lower eight bits	

Note:
QD/QP/QG series: When telemetry status frame 2 is not enabled, the telemetry data frequency is 50hz, and the frame 1 refresh rate is 50hz;When telemetry status frame 2 is enabled,the telemetry data frequency is 50hz, frame 1 refresh rate is 25hz,the frame 2 refresh rate is 25hz, alternately returning;The default state is not enabled.
QEC Series: When connected to pod equipment, the telemetry data frequency is 50hz,the frame 1 refresh rate is 25hz, and the frame 2 refresh rate is 25hz, alternately returning; When the pod equipment is not connected, the telemetry data frequency is 50hz, the frame 1 refresh rate is 0hz, and frame 2 refresh rate is 50hz.

6.3. Telemetry Status Frame S33

Table 4 Telemetry status frame S33

Byte No.	Content	Type	Value and range
1	Frame Header1	UINT8	0xAA
2	Frame Header2	UINT8	0x55
3	Frame length	UINT16	Length of the entire frame data
4			
5	Frame Identification Code	UINT8	0x33, QEC state frame
6	work pattern	UINT8	Bit0: Target detection enabled (0: Off 1: On) Bit1: Everything recognition enabled (0: off 1: on) Bit2: Template matching enabled (0: Off 1: On)
7	continue to have	UINT8	
8	System temperature	INT8	LSB=1°C, -128 ~ 127
9	continue to have	UINT8	
10	continue to have	UINT8	
11	continue to have	UINT8	
12	continue to have	UINT8	
13	continue to have	UINT8	
14	continue to have	UINT8	
15	continue to have	UINT8	
16	continue to have	UINT8	
17	Frame end 1	UINT8	0xAA
18	Frame tail 2	UINT8	0x55
19	Checksum	UINT8	Sum all the previous bytes and take the lower eight bits
Note: Only QEC series is supported. Active return is enabled by default and the frequency is 5hz.			

6.4. Telemetry Status Frame S34

Table 5 Telemetry status frame S34

Byte No.	Content		Type	Value and range
1	Frame Header1		UINT8	0xAA
2	Frame Header2		UINT8	0x55
3	Frame length		UINT16	Length of the entire frame data
4				
5	Frame ID		UINT8	0x34, Threat Target Status Frame
6	Image Frame ID		UINT32	Image Frame ID
7				
8				
9				
10	Timestamp		UINT64	Current external synchronized timestamp, LSB = 1ms
11				
12				
13				
14				
15				
16				
17				
18	Threat	Target Alarm Flag	UINT8	Bit0: TVU Alarm 0x00: Invalid; 0x01: Valid; Bit1: IRU Alarm 0x00: Invalid; 0x01: Valid; Bit2: TVD Alarm 0x00: Invalid; 0x01: Valid; Bit3: IRD Alarm 0x00: Invalid; 0x01: Valid; Bits4–7: Reserved
19	Target Count		UINT8	Maximum number: 32
20	Target 1	Type	UINT8	Bits 0~3: Target Category 0x00: Aircraft 0x01: Vessel 0x02: Vehicle 0x03: Person Bits 4~6: Target Source 0x00: Main Image 0x01: TV1 0x02: IR1 0x03: TV2

				0x04: IR2 0xFF: Template Matching Result
21		ID	UINT8	LSB = 1
22		Confidence	UINT8	LSB = 0.00390625
23		Horizontal Coordinate	UINT16	LSB = 1 pixel, origin at top-left, positive towards bottom-right
24				
25		Vertical Coordinate	UINT16	LSB = 1 pixel, origin at top-left, positive towards bottom-right
26				
27		Width	UINT16	LSB = 1 pixel
28				
29		Height	UINT16	LSB = 1 pixel
30				
	...			
	Target n			
19+n*11	Frame Tail 1	UINT8	0xAA	
20+n*11	Frame Tail 2	UINT8	0x55	
21+n*11	Checksum	UINT8	Lower 8 bits of the sum of all preceding bytes	
SD Series: When telemetry status frame 2 is disabled, the telemetry data frequency is 50 Hz and frame 11 refresh rate is 50 Hz. When telemetry status frame 34 is enabled, the telemetry data frequency remains 50 Hz, with frame 11 refreshed at 25 Hz and frame 34 refreshed at 25 Hz alternately. The default state is disabled.				

6.5. Query Device Version Information Frame QS0

Table 6 Device Version Information Frame QS0

Byte No.	Content	Type	Value and range
1	Frame header 1	UINT8	0xAA
2	Frame header 2	UINT8	0x55
3	Frame length	UINT8	Total frame data length
4	Frame identification code	UINT8	0xE0: Device information frame
5	Device Model [20]	CHAR	Device model, corresponding ASCII characters e.g. QP-125S QP-125S-0000
...			
24			
25	Device Serial	CHAR	Device serial number, corresponding

...	No. [9]		ASCII characters e.g. 001-0001
33			
34	Software Version [50]	CHAR	Device software version, corresponding ASCII characters e.g.: HV1.0.0_SV1.0_1.0.0
...			
83			
84	Checksum	UINT8	Lower 8 bits of the sum of all preceding bytes

6.6. Query the device network information frame QS1

Table 7 Device network information frame QS1

Byte No.	Content	Type	Values and ranges
1	Frame header 1	UINT8	0xAA
2	Frame header 2	UINT8	0x55
3	Frame length	UINT8	Length of the entire frame
4	Frame identification code	UINT8	0xE1: Device information frame
5	IP address	CHAR	An IPv4 address, corresponding to the ASCII of the character e.g. 192.168.1.100
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25	subnet mask	-	Subnet mask, corresponding character ASCII
26			

27			e.g.: 255.255.255.0
28			
29			
30			
31			
32			
33			
34			
35			
36			
37			
38			
39			
40			
41			
42			
43			
44			
45	default gateway	-	Default gateway, corresponding to the ASCII of the character e.g. 192.168.1.1
46			
47			
48			
49			
50			
51			
52			
53			
54			
55			
56			
57			
58			
59			
60			
61			
62			
63			
64			
65	Checksum	UINT8	Sum all the previous bytes and take the lower eight bits

6.7. Query the laser encoded information frame QS2

Table 8 Queries the laser code information frame QS2

Byte No.	Content	Type	Values and ranges
1	Frame header 1	UINT8	0xAA
2	Frame header 2	UINT8	0x55
3	Frame length	UINT8	Length of the entire frame
4	Frame Identification Code	UINT8	0xE2, laser encoded information frame
5	Coding value	UINT16	0x01: Laser fixed code 1; 0x02: Laser fixed code 2; ... 0x101: Laser custom code 1; 0x102: Laser custom code 2; ...
6			
7	Coding cycle	UINT16	LSB=0.01ms
8	continue to have	UINT8	
9	continue to have	UINT8	
10	continue to have	UINT8	
11	continue to have	UINT8	
12	continue to have	UINT8	
13	Checksum	UINT8	Sum all the previous bytes and take the lower eight bits

6.8. Query the laser compensation information frame QS3

Table 9 Queries the laser code information frame QS3

Byte No.	Content	Type	Values and ranges
1	Frame header 1	UINT8	0xAA
2	Frame header 2	UINT8	0x55
3	Frame length	UINT8	Length of the entire frame
4	Frame identification code	UINT8	0xE3, laser compensation information frame
5	Compensation value 0	INT16	Compensation values0,LSB=0.1m
6			
7	Compensation value 1	INT16	Compensation values1,LSB=0.1m
8			
9	Compensation value 2	INT16	Compensation values2,LSB=0.1m
10			
11	Compensation value 3	INT16	Compensation values3,LSB=0.1m
12			
13	Compensation value 4	INT16	Compensation values4,LSB=0.1m
14			
15	Compensation value 5	INT16	Compensation values5,LSB=0.1m
16			
17	Compensation value 6	INT16	Compensation values6,LSB=0.1m
18			
19	Compensation value 7	INT16	Compensation values7,LSB=0.1m
20			
21	Compensation value 8	INT16	Compensation values8,LSB=0.1m
22			
23	Compensation value 9	INT16	Compensation values9,LSB=0.1m
24			
25	Checksum	UINT8	Sum all the previous bytes and take the lower eight bits

6.9. Query Custom OSD Information Frame QS4

Table 10 QS4 Custom OSD Information Query Frame

Byte No.	Content	Type	Values and ranges
1	Frame header 1	UINT8	0xAA
2	Frame header 2	UINT8	0x55
3	Frame length	UINT8	Length of the entire frame
4	Frame identification code	UINT8	0xE4, Custom OSD Information Frame
5	Status	UINT8	Bit 0: Switch – 0 = Off, 1 = On
			Bits 2~7: Reserved
6	Horizontal Coordinate	UINT16	Horizontal Coordinate: Origin (0) at the top-left corner, positive to the right
7			
8	Vertical Coordinate	UINT16	Vertical Coordinate: Origin (0) at the top-left corner, positive downward
9			
10	Color A	UINT8	
11	Color R	UINT8	
12	Color G	UINT8	
13	Color B	UINT8	
N	Custom Data	UINT8	UTF-8 Encoded Characters
N+1	Checksum	UINT8	Sum all preceding bytes and take the lower 8 bits